CCMI - PhD Training on the Interface of Data, Software, and Computation

Timo Betcke, UCL

A career on the fence

The future needs fence sitters

Fusion Computing Lab: engaging supercomputers to test the "untestable"

UCL Advanced Research Computing Centre

What is CCMI?

- Joint Doctoral Training Centre between UCL and Imperial College
- Over £13m total investment across EPSRC, UCL, Imperial, and partners
- Funding to train over 70 PhD students across 5 cohorts
- Led by UCL Advanced Research Computing

The three pillars

Supported Research

CCMI specific training

Collaborative Working Groups

Software Journey

Termly Software Weeks

Admissions

- We want students that are passionate about our vision, not just students with the best transcripts
- Students don't need to be expert coders but they need to be willing to become one
- EDI should be at the heart of all our admissions decisions

Online Application

Interview

Project Matching

Online Application

The interview

Gender balance

Project Matching

Outreach activities

Are we AI yet?

- We wrote the CDT proposal before the Claude Caude big bang.
- Almost all our students have AI subscriptions. Most use coding agents.
- We have dome one agentic coding workshop so far
- In discussions with partners on regular agentic training events.

Collaborations

A snapshot of partners

Visit to STFC Rutherford Appleton Labs

How to engage

- Talk to us about collaborative software projects
- Co-Funding of studentships (a full studentship is around £140k)
- If you are a UCL/Imperial advisor simply talk to us to advertise relevant projects
- Come and visit us in our software weeks and show off your work to our students

A few final thoughts

What works

- Students across two universities and nine departments code and collaborate together
- We have extremely good application numbers, including very strong home student application rates
- A lot of good will at UCL to make a CDT work that is led by a non-academic department

Student voices

What are the challenges?

- Developing software projects with enough critical support by students is challenging. Will require multi-cohort approaches.
- Gender balance in year 2 better than year 1 but not yet where we want it to be.
- Still institutional cultural challenges to PhD training that is not led by an academic department.

Many thanks to

Collaborative Computational Modelling at the Interface

UCL

IMPERIAL

UK Research and Innovation

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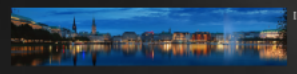

I grew up in Hamburg, Germany

I studied computational engineering at the Technical University Hamburg

DPhil under Nick Trefethen in the Oxford Computing Lab

Stations in Braunschweig, Manchester, and Reading

- Since 2011 at UCL Department of Mathematics
- Professor since 2018
- Since 2022 also Academic Fellow at UCL Advanced Research Computing Centre (ARC)

Fence sitters drive science

Good software makes good science
...good science makes good software.
Why distinguish the two?

The future needs fence sitters

Fusion Computing Lab: engaging supercomputers to test the "untestable"

UKAEA's Fusion Computing Lab uses supercomputing, AI and digital twins to accelerate safe, scalable design of future fusion energy power plants

https://www.ukaea.org/case-studies/fusion-computing-lab-first-digital-twin-case-study/

UCL Advanced Research Computing Centre

nature

Why science needs more research software engineers

Ten years after their profession got its name, research software engineers seek to swell their ranks.

- On the interface of professional service and digital research
- Staff have PS contracts but same opportunities for promotion, grants, and research supervision as academic departments

Advanced Research Computing Centre

ARC is UCL's research, innovation and service centre for the tools, practices and systems that enable computational science and digital scholarship.

We are an innovative centre with both professional services and academic missions:

- We provide advanced, reliable and secure digital research infrastructure to research groups in UCL and beyond.
- We are a laboratory for research, teaching and innovation in compute, data and software intensive research methods.

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BLAS and Lapack

(Almost) every numerical software needs basic matrix operations and decompositions

BLAS (Basic Linear Algebra Subprograms, 1978) defines the interfaces. Efficient vendor libraries implement them.

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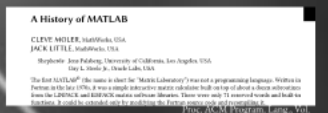
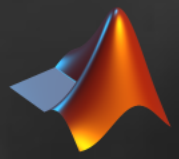


Proc. ACM Program. Lang., Vol. 4



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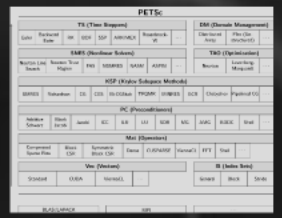

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deal.II - Differential Equations Analysis Library

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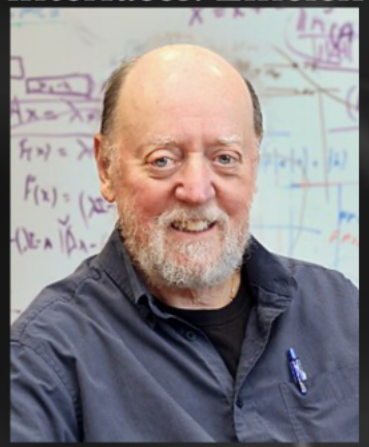
BLAS and Lapack

(Almost) every numerical software needs basic matrix operations and decompositions

Level 3 BLAS: matrix-matrix, $O(n^3)$ operations

types	name	(options	size	arguments	)	description
s, d, c, z	gemm	(transA, transB,	m, n, k,	alpha, A, ldA, B, ldB, beta, C, ldC	)	general matrix-matrix multiply
s, d, c, z	gemmtr	(uplo, transA, transB,	m, n, k,	alpha, A, ldA, B, ldB, beta, C, ldC	)	general matrix-matrix multiply
s, d, c, z	symm	(side, uplo,	m, n,	alpha, A, ldA, B, ldB, beta, C, ldC	)	symmetric matrix-matrix mul.
c, z	hemm	(side, uplo,	m, n,	alpha, A, ldA, B, ldB, beta, C, ldC	)	Hermitian matrix-matrix mul.
s, d, c, z	trmm	(side, uplo, transA, diag,	m, n,	alpha, A, ldA, B, ldB	)	triangular matrix-matrix mul.
s, d, c, z	trsm	(side, uplo, transA, diag,	m, n,	alpha, A, ldA, B, ldB	)	triangular solve matrix
s, d, c, z	syrk	(uplo, trans,	n, k,	alpha, A, ldA,	beta, C, ldC)	symmetric rank-k update
c, z	herk	(uplo, trans,	n, k,	alpha, A, ldA,	beta, C, ldC)	Hermitian rank-k update
s, d, c, z	syr2k	(uplo, trans,	n, k,	alpha, A, ldA, B, ldB,	beta, C, ldC)	symmetric rank-$2k$ update
c, z	her2k	(uplo, trans,	n, k,	alpha, A, ldA, B, ldB,	beta, C, ldC)	Hermitian rank-$2k$ update

BLAS (Basic Linear Algebra Subprograms, 1978) defines the interfaces. Efficient vendor libraries implement them.



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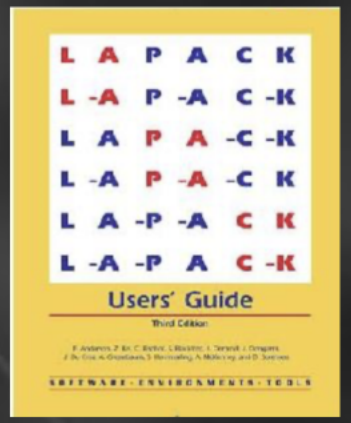

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Matlab

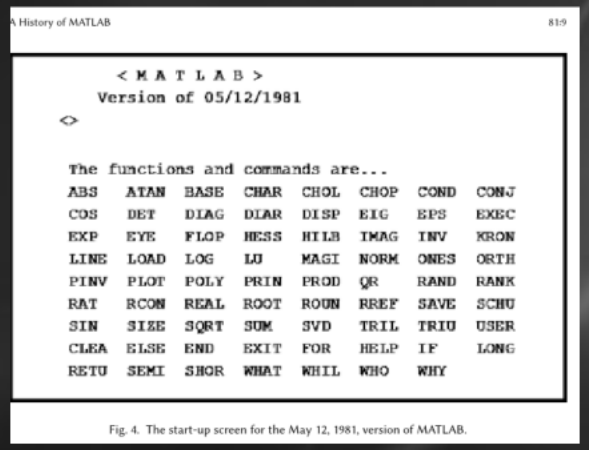

Fig. 4. The start-up screen for the May 12, 1981, version of MATLAB.

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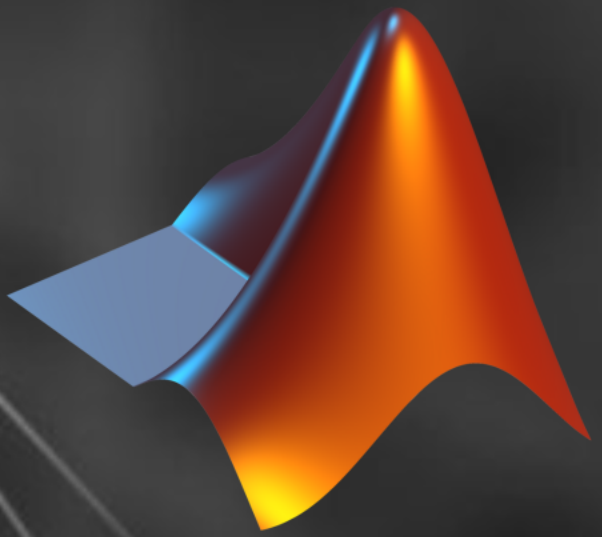


A History of MATLAB

CLEVE MOLER, MathWorks, USA
JACK LITTLE, MathWorks, USA

Shepherds: Jens Palsberg, University of California, Los Angeles, USA
Guy L. Steele Jr., Oracle Labs, USA

The first MATLAB® (the name is short for "Matrix Laboratory") was not a programming language. Written in Fortran in the late 1970s, it was a simple interactive matrix calculator built on top of about a dozen subroutines from the LINPACK and EISPACK matrix software libraries. There were only 71 reserved words and built-in functions. It could be extended only by modifying the Fortran source code and recompiling it.

Proc. ACM Program. Lang., Vol. 4

APPROXIMATIONS AND BOUNDS FOR EIGENVALUES OF ELLIPTIC OPERATORS*

L. FOX†, P. HENRICI‡ AND C. MOLER§

1. Introduction. Let G be a bounded, n-dimensional domain with boundary Γ. Let $p(x)$ and $q_i(x)$, $i = 1, \cdots, n$, be positive functions defined on $\bar{G}$ and let D denote the linear, self-adjoint differential operator defined by

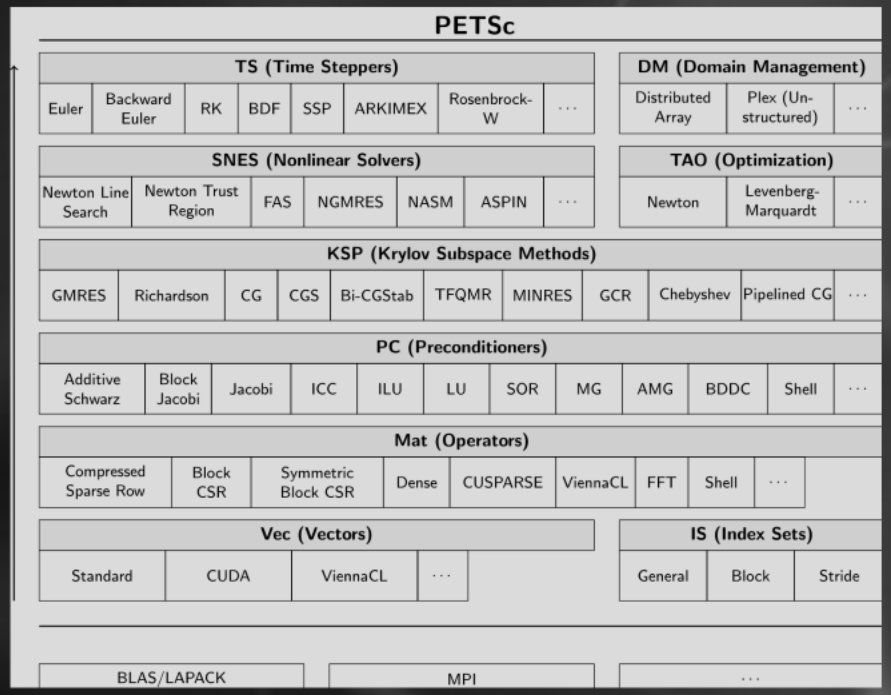
$$Du = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(q_i \frac{\partial u}{\partial x_i}\right).$$

Cleve Moler, Chief Mathematician

We are saddened to share that Cleve Moler passed away on May 20, 2026, at the age of 86 at his home surrounded by his family. Cleve was chief mathematician and cofounder of MathWorks and the author of the first version of MATLAB.

In his early years, he was a professor of math and computer science for almost 20 years at the University of Michigan, Stanford University, and the University of New Mexico. During this time he was known for being one of the authors of LINPACK and EISPACK, two foundational Fortran libraries for numerical computing. One popular paper of his is "Nineteen Dubious Ways to Compute the Exponential of a Matrix."

PETSc - Portable, Extensible Toolkit for Scientific Computing



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UKAEA’s Fusion Computing Lab uses supercomputing, AI and digital twins to accelerate safe, scalable design of future fusion energy power plants

https://www.ukaea.org/case-studies/fusion-computing-lab-first-digital-twin-case-study/

UCL Advanced Research Computing Centre

nature

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CAREER Q&A | 31 May 2022

Why science needs more research software engineers

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By Chris Woolston

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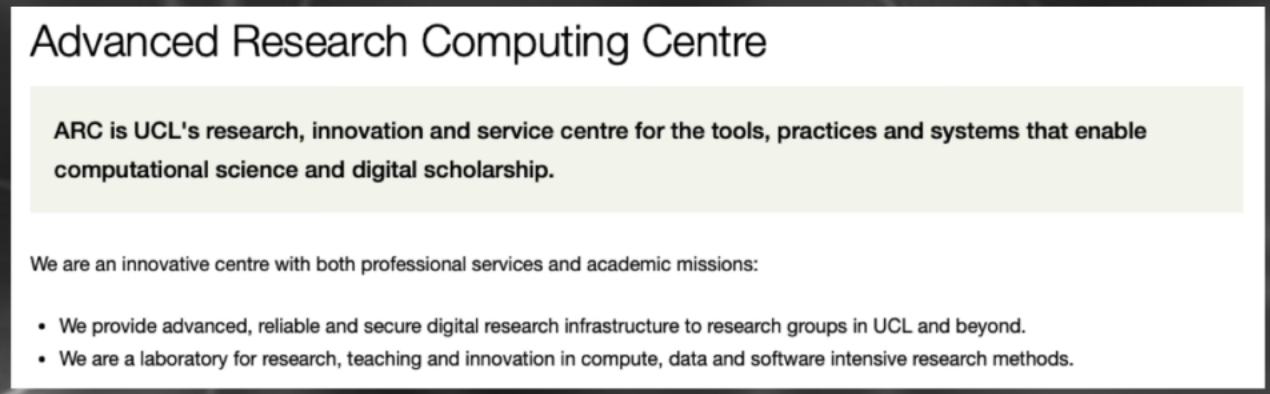

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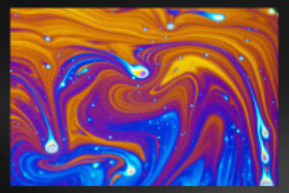
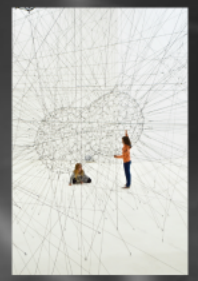


Computational Simulation

Research Software Engineering



Data Sciences

Supported Research

Research Priorities

- Mathematical Foundations
- Computational Algorithms
- Scientific Research Software Engineering
- Bridging the gap to applications

Imperial Aeronautics and Design Engineering to join in 2026

Our students are pursuing research in nine departments across UCL and Imperial.

- Imperial Departments: Mathematics, Computing, Chemistry
- UCL Departments: Mathematics, Statistics, Computer Science, Advanced Research Computing, Chemistry, Division of Medicine

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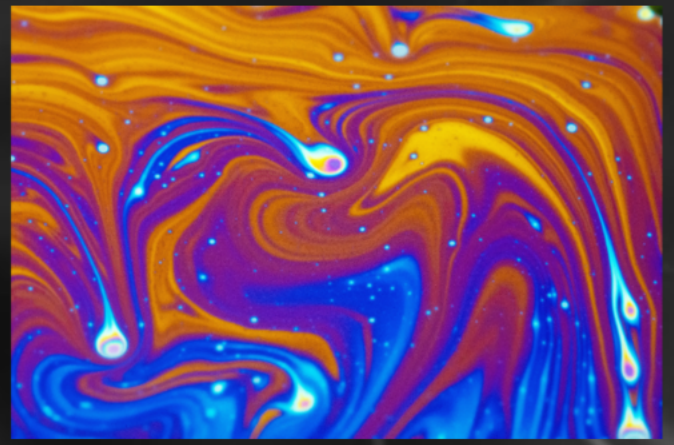

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27. Tell us about your academic and professional background (max 750 characters, incl spaces)

For example, I have a BSc in **x** and an MSc in **y**. I did the following courses that are relevant... I did a research project in **z**. I worked in Industry as **w** for **n** months/years etc *

Please enter at most 750 characters

28. Select which of these aspirations best describe the type of skills that you would like to acquire during your training and how you would like to use them. We recognise that you might align with more than one option, but please select the option you most closely identify with. This will broadly specify the type of PhD project you could be interested in. *

- A. I want to develop and use computational models to better understand phenomena in the world around us.
- B. I want to understand and extend the mathematical foundations underpinning computational models (accuracy, stability, tractability, etc)
- C. I want to develop tools for blending computational models and data, to make inferences and predictions.
- D. I want to design, analyse and implement new algorithms enabling computational models to run efficiently and scalably on modern hardware.
- E. I want to work on principled and efficient solution of complex real world problems.

29. What aspects of your experience motivate you and will support you in being successful in the aspiration you selected in the question above? (max 750 characters, incl spaces)

For example: when I did course **x**, I realised how much I like working on **y**. Or, the project I did in **z** has given me a solid background in **w** that is relevant because...; Or, when I was working as **h**, I realised that I want a future in **q**. *

30. What aspect of the CCMI PhD experience are you most looking forward to, and why? (max 750 characters, incl spaces) *

31. What 3 things, in your career thus far, are you most proud of, and why? (max 1000 characters, incl spaces)

For example: I was very proud of obtaining a high mark on course **x**; winning award **y**; being involved in project **z**, publishing research paper on **h**; being successful in job **w** achieving **t**; Producing software **z**; Overcoming challenge **q**; Solving problem **n** etc. *

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34. Is there anything else you would like to add for us to consider whilst reviewing your application? For example, career breaks or challenges that affected your professional development? This section is optional. (max 750 characters, incl spaces)

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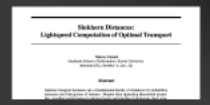

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Outreach activities

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Online Application

27. Tell us about your academic and professional background (max 750 characters, incl spaces)

    For example, I have a BSc in **x** and an MSc in **y**. I did the following courses that are relevant... I did a research project in **z**. I worked in Industry as **w** for **n** months/years etc *

    Please enter at most 750 characters

28. Select which of these aspirations best describe the type of skills that you would like to acquire during your training and how you would like to use them. We recognise that you might align with more than one option, but please select the option you most closely identify with. This will broadly specify the type of PhD project you could be interested in. *

    - A. I want to develop and use computational models to better understand phenomena in the world around us.
    - B. I want to understand and extend the mathematical foundations underpinning computational models (accuracy, stability, tractability, etc)
    - C. I want to develop tools for blending computational models and data, to make inferences and predictions.
    - D. I want to design, analyse and implement new algorithms enabling computational models to run efficiently and scalably on modern hardware.
    - E. I want to work on principled and efficient solution of complex real world problems.

29. What aspects of your experience motivate you and will support you in being successful in the aspiration you selected in the question above? (max 750 characters, incl spaces)

    For example: when I did course **x**, I realised how much I like working on **y**; Or, the project I did in **z** has given me a solid background in **w** that is relevant because....; Or, when I was working as **h,** I realised that I want a future in **q.** *

30. What aspect of the CCMI PhD experience are you most looking forward to, and why? (max 750 characters, incl spaces) *

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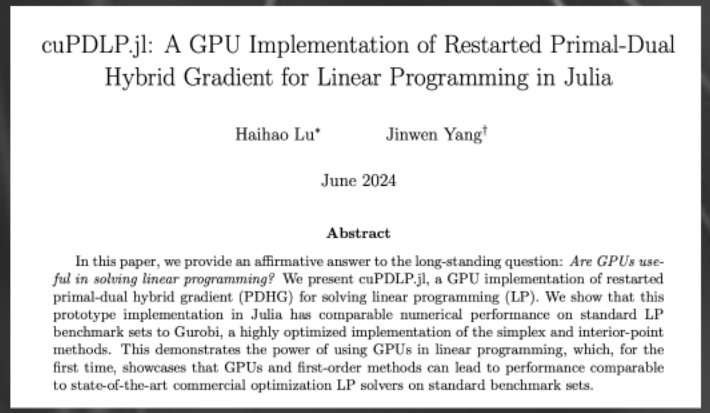


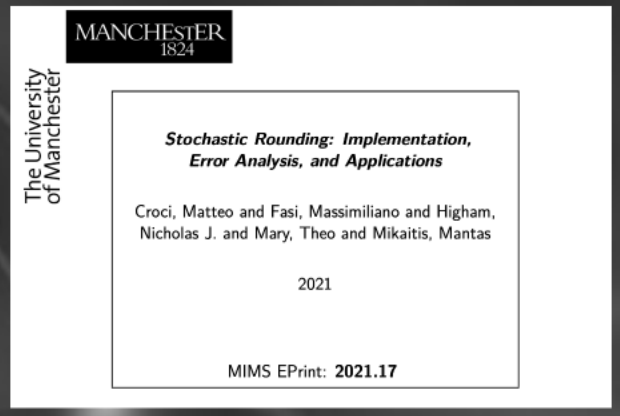
cuPDLP.jl: A GPU Implementation of Restarted Primal-Dual Hybrid Gradient for Linear Programming in Julia

Haihao Lu* Jinwen Yang†

June 2024

Abstract

In this paper, we provide an affirmative answer to the long-standing question: *Are GPUs useful in solving linear programming?* We present cuPDLP.jl, a GPU implementation of restarted primal-dual hybrid gradient (PDHG) for solving linear programming (LP). We show that this prototype implementation in Julia has comparable numerical performance on standard LP benchmark sets to Gurobi, a highly optimized implementation of the simplex and interior-point methods. This demonstrates the power of using GPUs in linear programming, which, for the first time, showcases that GPUs and first-order methods can lead to performance comparable to state-of-the-art commercial optimization LP solvers on standard benchmark sets.



MANCHESTER 1824

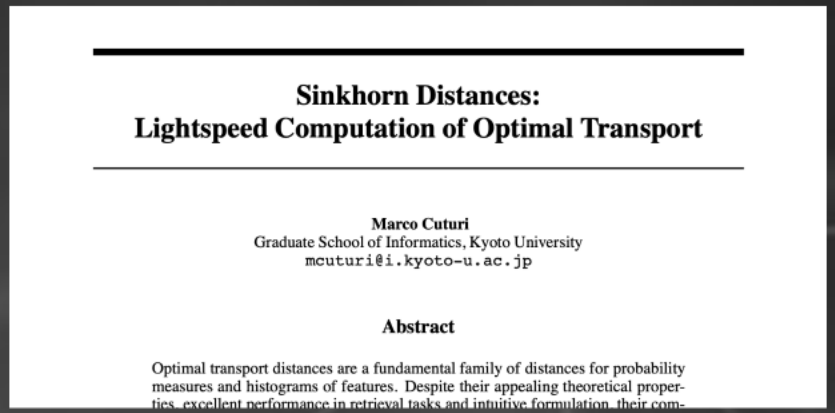
The University of Manchester

Stochastic Rounding: Implementation, Error Analysis, and Applications

Croci, Matteo and Fasi, Massimiliano and Higham, Nicholas J. and Mary, Theo and Mikaitis, Mantas

2021

MIMS EPrint: **2021.17**



Sinkhorn Distances: Lightspeed Computation of Optimal Transport

Marco Cuturi
Graduate School of Informatics, Kyoto University
mcuturi@i.kyoto-u.ac.jp

Abstract

Optimal transport distances are a fundamental family of distances for probability measures and histograms of features. Despite their appealing theoretical proper-
ties, excellent performance in retrieval tasks and intuitive formulation, their com-

Project Matching

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The Mary Lister McCammon Summer Research Fellowship

CCMI specific training

Collaborative Working Groups

- 7 Training workshops across the first two years
- Each training workshop designed specifically for the CDT, combining lecture style content, student presentations and software work

Year 1

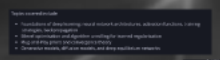
- Scientific Computing
- Research Software Engineering
- Data-Intensive Computations
- PDE Discretisations
- Deep Learning

Year 2

High-Dimensional Problems
Responsible computational modelling

Programming Models for Scientific Computing

Deep Learning



- Half-Day Theory followed by afternoon coding challenge

Software Journey

- Ongoing Software Projects
- Termly software weeks for all students
- Talks by companies

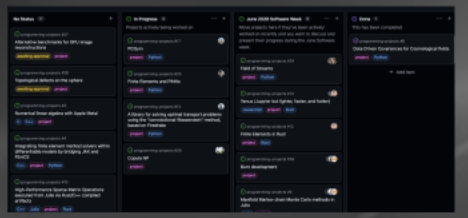
Termly Software Weeks

- Mixture of hackathons, software discussions and external visits

Some highlights

- Internal full day sandpit to allocate new projects
- Visit by Graphcore to talk about AI hardware/software
- One day training workshop by AWS on agentic coding

Software Projects



ND - A numerical discretisation framework for Rust

Machine Learning in Rust with Burn

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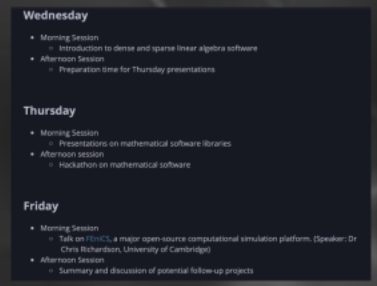
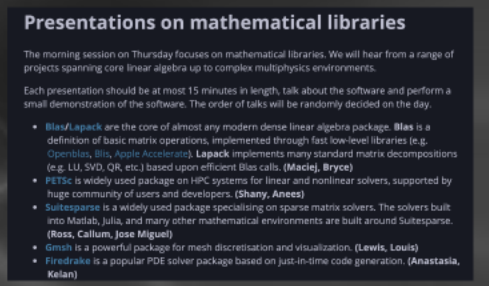
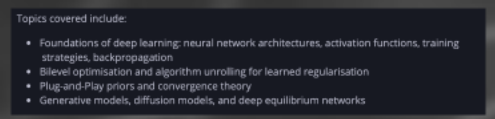




Deep Learning



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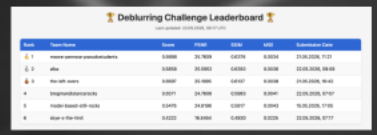

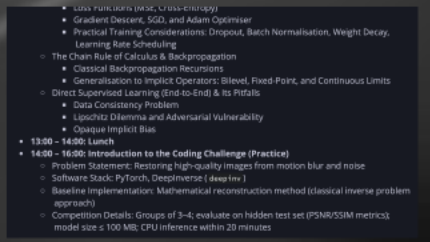

Programming Models for Scientific Computing

Wednesday

- Morning Session
  - Introduction to dense and sparse linear algebra software
- Afternoon Session
  - Preparation time for Thursday presentations

Thursday

- Morning Session
  - Presentations on mathematical software libraries
- Afternoon session
  - Hackathon on mathematical software

Friday

- Morning Session
  - Talk on FEniCS, a major open-source computational simulation platform. (Speaker: Dr Chris Richardson, University of Cambridge)
- Afternoon Session
  - Summary and discussion of potential follow-up projects

Presentations on mathematical libraries

The morning session on Thursday focuses on mathematical libraries. We will hear from a range of projects spanning core linear algebra up to complex multiphysics environments.

Each presentation should be at most 15 minutes in length, talk about the software and perform a small demonstration of the software. The order of talks will be randomly decided on the day.

- **Blas/Lapack** are the core of almost any modern dense linear algebra package. **Blas** is a definition of basic matrix operations, implemented through fast low-level libraries (e.g. Openblas, Blis, Apple Accelerate). **Lapack** implements many standard matrix decompositions (e.g. LU, SVD, QR, etc.) based upon efficient Blas calls. **(Maciej, Bryce)**
- **PETSc** is widely used package on HPC systems for linear and nonlinear solvers, supported by huge community of users and developers. **(Shany, Anees)**
- **Suitesparse** is a widely used package specialising on sparse matrix solvers. The solvers built into Matlab, Julia, and many other mathematical environments are built around Suitesparse. **(Ross, Callum, Jose Miguel)**
- **Gmsh** is a powerful package for mesh discretisation and visualization. **(Lewis, Louis)**
- **Firedrake** is a popular PDE solver package based on just-in-time code generation. **(Anastasia, Kelan)**

Deep Learning

Topics covered include:

- Foundations of deep learning: neural network architectures, activation functions, training strategies, backpropagation
- Bilevel optimisation and algorithm unrolling for learned regularisation
- Plug-and-Play priors and convergence theory
- Generative models, diffusion models, and deep equilibrium networks

- Half-Day Theory followed by afternoon coding challenge

🏆 Deblurring Challenge Leaderboard 🏆

Last updated: 22.05.2026, 09:17 UTC

Rank	Team Name	Score	PSNR	SSIM	MSE	Submission Date
🥇 1	moore-penrose-pseudostudents	0.5888	25.7609	0.6378	0.0034	21.05.2026, 11:21
🥈 2	alba	0.5858	25.5952	0.6350	0.0036	22.05.2026, 09:09
🥉 3	the-left-overs	0.5697	25.1885	0.6137	0.0038	21.05.2026, 16:42
4	bregmandistancerocks	0.5571	24.7606	0.5983	0.0041	22.05.2026, 07:57
5	model-based-still-rocks	0.5476	24.8198	0.5817	0.0043	15.05.2026, 17:05
6	skye-s-the-limit	0.4222	16.5454	0.4830	0.0225	22.05.2026, 07:17

    - Loss Functions (MSE, Cross-Entropy)
    - Gradient Descent, SGD, and Adam Optimiser
    - Practical Training Considerations: Dropout, Batch Normalisation, Weight Decay, Learning Rate Scheduling
  - The Chain Rule of Calculus & Backpropagation
    - Classical Backpropagation Recursions
    - Generalisation to Implicit Operators: Bilevel, Fixed-Point, and Continuous Limits
  - Direct Supervised Learning (End-to-End) & Its Pitfalls
    - Data Consistency Problem
    - Lipschitz Dilemma and Adversarial Vulnerability
    - Opaque Implicit Bias
- **13:00 – 14:00: Lunch**
- **14:00 – 16:00: Introduction to the Coding Challenge (Practice)**
  - Problem Statement: Restoring high-quality images from motion blur and noise
  - Software Stack: PyTorch, DeepInverse (`deepinv`)
  - Baseline Implementation: Mathematical reconstruction method (classical inverse problem approach)
  - Competition Details: Groups of 3–4; evaluate on hidden test set (PSNR/SSIM metrics); model size ≤ 100 MB; CPU inference within 20 minutes

Software Journey

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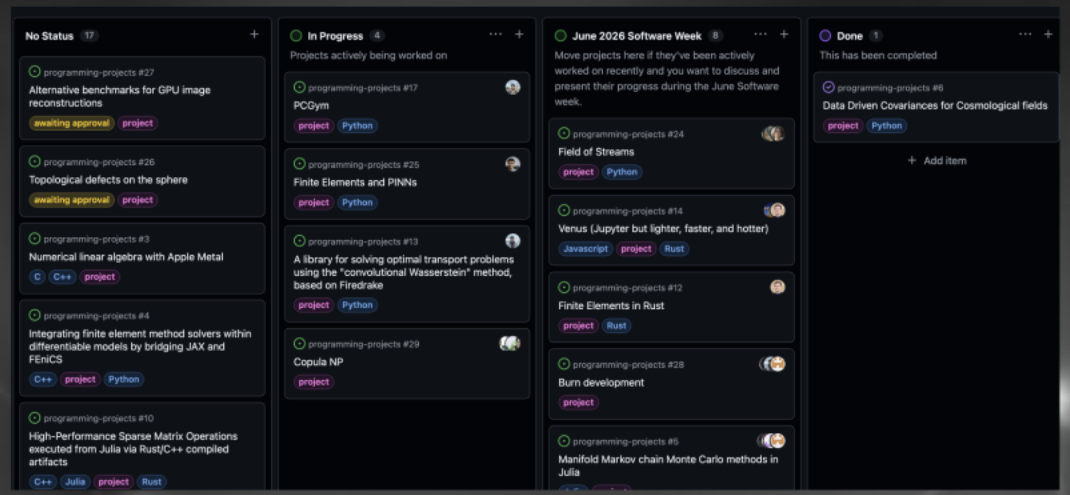
- Termly software weeks for all students
- Talks by companies

Software Projects



Termly Software Weeks

- Mixture of hackathons, software discussions and external visits

Some highlights

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- Visit by Graphcore to talk about AI hardware/software
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ND - A numerical discretisation framework for Rust

- Large scale numerical discretisation in Rust
- Have already run on Archer2 with hundreds of million mesh elements

Machine Learning in Rust with Burn

First PR to add functionality to backpropagation graphs

Anees Hussain

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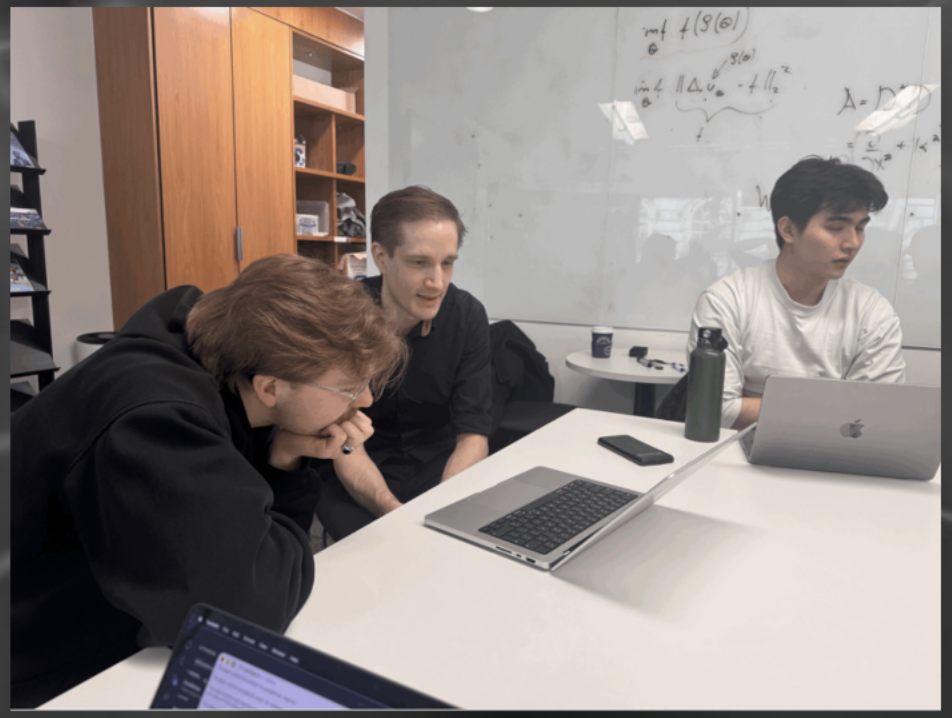
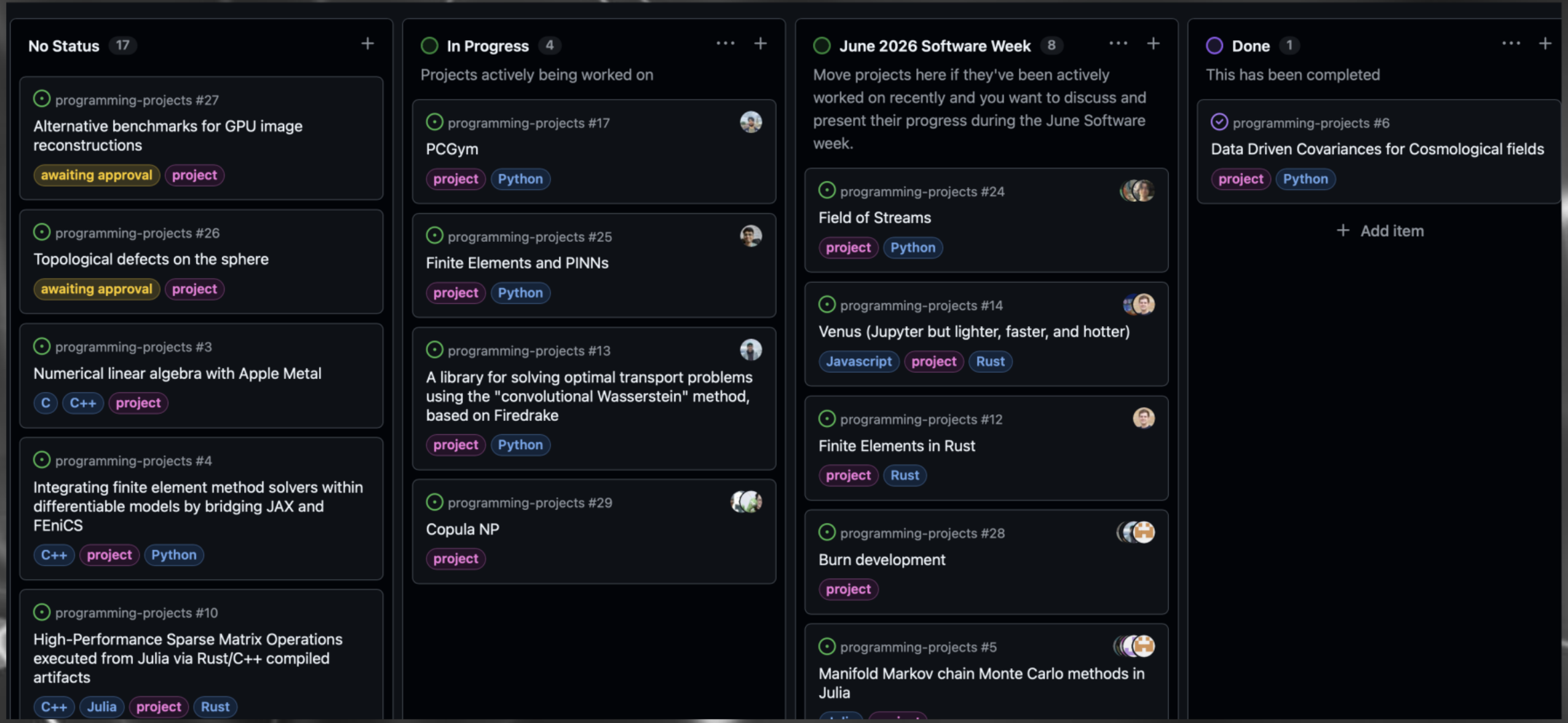


Software Projects



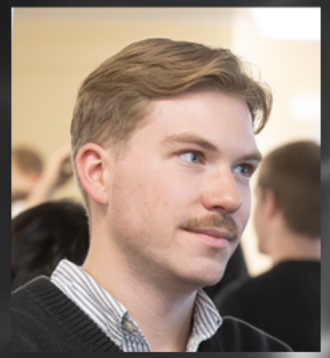
ND - A numerical discretisation framework for Rust

```rust
pub trait FiniteElement {
    type T;
    type CellType;

    fn cell_type(&self) -> Self::CellType;

    fn dim(&self) -> usize;

    fn value_shape(&self) -> &[usize];

    fn value_size(&self) -> usize;

    fn tabulate(
        &self,
        points: &Array2D<Self::T>,
        nderivs: usize,
        data: &mut Array4D<Self::T>,
    );
```

- Large scale numerical discretisation in Rust
- Have already run on Archer2 with hundreds of million mesh elements

Matthew Scroggs

Shany Neeman



Maciej Kaczorek

ndelement and ndmesh: numerical discretisation in Rust

Timo Betcke[1] **and Matthew W. Scroggs**[2]

1 Department of Mathematics, University College London **2** Advanced Research Computing Centre, University College London

Summary

When working with partial differential equations (PDEs) that arise from problems in science or engineering, it is common to use numerical methods to obtain approximate solutions. Many numerical methods for PDEs, such as the finite element method (Ciarlet, 1978; Johnson, 2009) and boundary element method (Steinbach, 2008), involve the discretisation of a domain into a mesh of polygonal or polyhedral cells, and the definition of a set of (usually) polynomial basis functions on each cell. A library implementing one of these methods will typically be built around two core pieces of functionality: the creation and storage of a mesh, and the definition of basis functions on a cell.

Machine Learning in Rust with Burn

First PR to add functionality to backpropagation graphs

Anees Hussain

Retain graph feature for partial derivatives #5043

Draft AHA-HH wants to merge 9 commits into `tracel-ai:main` from `AHA-HH:feat/retaingraph`

Conversation 0 · Commits 9 · Checks 9 · Files changed 24

AHA-HH commented 2 weeks ago

Retain graph feature for partial derivatives

If there is something off with the PR, apologies in advance, I'm new to this.

Checklist

- [x] Confirmed that `cargo run-checks` command has been executed.*
- [x] Made sure the book is up to date with changes in this PR.**

*cargo run-checks fails on the LibTorch backend. This is an environment issue unrelated to the changes in this PR. All autodiff and backend tests pass when run directly with cargo test except for the rfft tests due to ndarray not being supported.
**Added a Retaining the Computational Graph section to burn-book/src/building-blocks/autodiff.md documenting backward_retain(), its use case and the memory tradeoff versus standard backward().

Related Issues/PRs

Solves #1942 #121

Changes

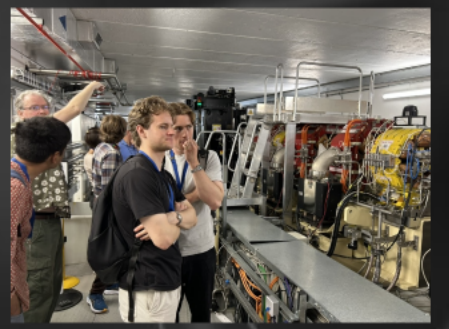
Burn's computational graph is stored as a HashMap<NodeId, StepBoxed> that ceases to exist after a single .backward() call. The proposed solution adds backward_retain() equivalent to PyTorch's loss.backward(retain_graph=True) which

Collaborations

A snapshot of partners

Visit to STFC Rutherford Appleton Labs

- STFC funds two studentships aligned with the CDT
- In May the whole CDT visited the Scientific Computing Group at RAL
- Establish new collaborations on software and projects



How to engage

- Talk to us about collaborative software projects
- Co-funding of studentships (a full studentship is around £140k)
- If you are a UCL/Imperial advisor simply talk to us to advertise relevant projects
- Come and visit us in our software weeks and show off your work to our students

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Are we AI yet?

- We wrote the CDT proposal before the Claude Caude big bang.
- Almost all our students have AI subscriptions. Most use coding agents.
- We have done one agentic coding workshop so far
- In discussions with partners on regular agentic training events.

A few final thoughts

What works

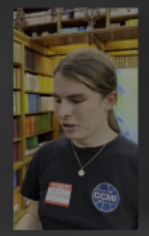
- Students across two universities and nine departments code and collaborate together
- We have extremely good application numbers, including very strong home student application rates
- A lot of good will at UCL to make a CDT work that is led by a non-academic department

What are the challenges?

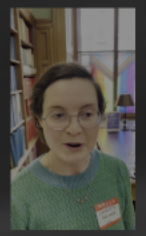
- Developing software projects with enough critical support by students is challenging. Will require multi-cohort approaches.
- Gender-balance in year 2 better than year 1 but not yet where we want it to be.
- Still institutional cultural challenges to PhD training that is not led by an academic department.

Student voices

Anees Hussain



Skye Purchase

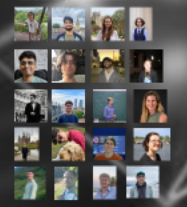


Charlotte Savage

Many thanks to

Our Leadership Team

...and all our postgraduate students



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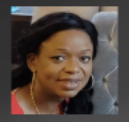
Overall Leadership

Timo Betcke
Lead

Colin Cotter
Co-Lead

Laura Beer
Planning and Operations


Lydia Noa
Imperial Manager

Matthew Scroggs
Teaching and Training

Research and Partnership

Marta Betcke
(UCL)

Ruth Misener
(Imperial)

Admissions

Hao Ni
(UCL)

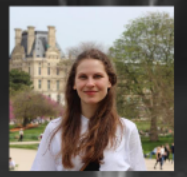
Vahid Shahrezaei
(Imperial)

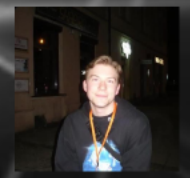
Training

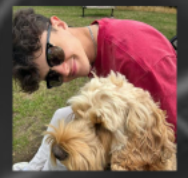
Serge Guillas
(UCL)

Dante Kalise
(Imperial)

...and all our year one students



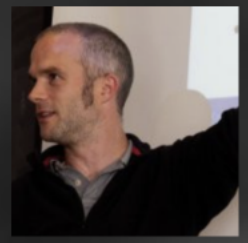




Our Leadership Team

Overall Leadership

Timo Betcke
Lead



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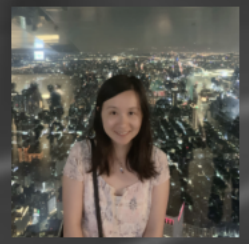

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